

Kapruka Flow AI

Kapruka Agent Challenge 2026

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Live demo: kapruka-flow-ai.vercel.app

Source: github.com/Thanuka9/kapruka-flow-AI

Tell me what you need. I'll build the cart.

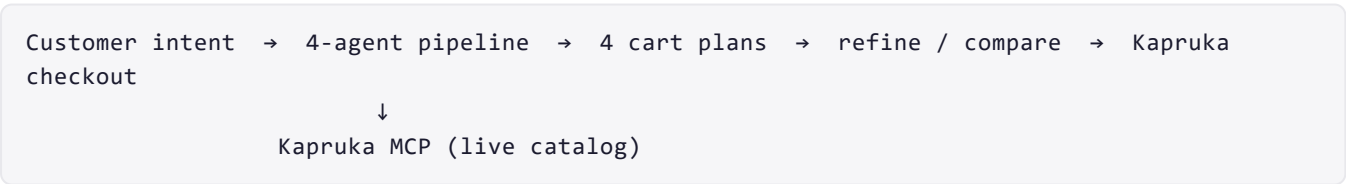
What this is

Kapruka Flow is an AI-first shopping experience built on the public [Kapruka MCP](#). Instead of browsing category by category, a customer describes what they want in natural language. The system turns that intent into **four budget-aware cart plans**, lets them compare and refine, then completes a **real guest checkout** on Kapruka.

Problem it solves

Typical e-commerce	Kapruka Flow
Search keywords, open many product pages	One sentence of intent
One cart, manual tweaks	Four optimized plans (Ideal, Cheaper, Premium, Fast)
Opaque recommendations	Visible MCP tool activity and curation story
English-only UX friction	English, Sinhala, and Tanglish (UI + prompts)

How it works (60-second overview)



1. **Customer** types or speaks a goal (e.g. "Amma birthday gift under 10000, deliver to Kandy tomorrow").
2. **Backend** runs a four-step agent pipeline against live MCP data.
3. **Frontend** shows the workspace animation, MCP activity feed, and four cart tabs.
4. **Customer** adjusts budget, chats with **Ruka** to refine, or compares versions.
5. **Checkout** validates stock and price via MCP, then returns a **Kapruka payment URL**.

No external LLM API key is required. Planning uses a **deterministic MCP-native engine** (intent parsing, relevance scoring, budget filling) so behavior is predictable and demo-safe.

The four-agent pipeline

Step	Agent	Role
1	Intent	Parse language, budget, city, date, occasion, recipient from the prompt
2	Catalog	Search and enrich products from Kapruka MCP
3	Delivery	Validate city and resolve delivery fee via MCP
4	Cart	Build four plan variants within budget using relevance scoring

Every MCP call is surfaced in the UI activity ticker so judges can see *what* the system did, not just the final cart.

What it can do

Shopping intelligence

- Multilingual intent: **English, Sinhala, Tanglish**
- Four cart strategies: **Ideal, Cheaper, Premium, Fast delivery**
- Instant **budget slider** rebalance (client-side)
- **Conversational refine** with Ruka (add/remove items, change city or budget)
- **Compare Versions** matrix and cart evolution timeline
- **Age-restricted** product deprioritization in ranking

Kapruka MCP integration (live)

- Product search and enrichment
- Category navigation (65+ categories)
- Delivery city validation and fees
- Stock and price check at checkout
- Guest order creation with payment link

Customer experience

- Kapruka-style header (search, categories, cart)
- Trending product cards on landing with **Add to cart**
- Sign in / register with order history for personalization
- Saved product bookmarks; shareable session links (?session=...)
- Localized UI in three languages

End-to-end flow (what judges should see)

#	Action	Expected result
1	Open the live demo	MCP healthy; landing shows search, trending products, commerce steps
2	Build a cart from a prompt	Workspace shows pipeline; four cart tabs populate
3	Switch plans / compare	Different items and prices per plan; comparison modal works
4	Refine via chat	e.g. "add chocolates" updates the active cart
5	Review & Checkout	Sender → Recipient → Delivery & gift → Review → Pay
6	Complete payment step	Kapruka MCP returns order reference and payment URL

Sample prompts:

- Amma birthday gift under 10000 deliver to Kandy tomorrow
- Ada Colombo ekata groceries 20000 ta adu (Tanglish)
- අම්මාට උපන්දින තෑග්ගක් කොළඹට (Sinhala)

Architecture (compact)

Layer	Technology	Responsibility
UI	Next.js 13, React	Intent canvas, cart workspace, checkout, localization
API proxy	Next.js /api/*	Forwards to backend with timeout and client IP
Orchestrator	FastAPI, Python 3.10	REST API, rate limits, sessions, checkout
Intelligence	mcp_intelligence.py	Intent parse, relevance score, cart optimization
Agents	agents.py	Four-step MCP pipeline
Data	SQLite (WAL)	Users, sessions, carts, orders, analytics
Catalog & orders	Kapruka MCP	All live product and checkout data

Design principles

1. **MCP-first** - Products and orders come from Kapruka MCP, not a static mock catalog.
2. **Transparent AI** - Tool names and pipeline steps are visible to the user.
3. **Plans, not a single guess** - Four carts respect different trade-offs (cost, quality, speed).
4. **Sri Lanka-ready** - Sinhala/Tanglish prompts and UI; LKR budgets; local delivery cities.
5. **Guest checkout** - No Kapruka account required; optional Flow account for history.

Security and reliability

- Passwords: PBKDF2-SHA256 with per-user salt

- CORS locked to deployed frontend origins
- Rate limits on intent, checkout, login, and search
- Checkout re-validates MCP price ($\pm 5\%$) and stock before order
- MCP calls: bounded timeout, retry with backoff
- Simulated checkout disabled in production

Quick links

Resource	URL
Live demo	https://kapruka-flow-ai.vercel.app
Repository	https://github.com/Thanuka9/kapruka-flow-AI
Kapruka MCP docs	https://mcp.kapruka.com/
Agent Challenge	https://www.kapruka.com/contactUs/agentChallenge.html

Suggested 5-minute judge walkthrough

1. **Landing** - Confirm trending products, Add to cart, and the six-step commerce strip.
2. **Build** - One multilingual prompt; watch MCP activity.
3. **Compare** - Switch Ideal / Cheaper / Premium / Fast; open Compare Versions.
4. **Refine** - Ask Ruka to add or remove an item.
5. **Checkout** - Fill sender, recipient, delivery city/date, gift message; proceed to Kapruka pay link.

Built for the Kapruka Agent Challenge 2026 using the public Kapruka MCP - no proprietary API key required.